

# Statistical Foundations of Prior-Data Fitted Networks [Nagler '23]

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Group 4



TabPFN predicts on a fresh table in one forward pass, no inference-time training.

A PFN trained on  $\leq 1000$  rows keeps improving past 1000. Why?

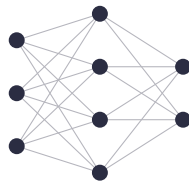
- PFNs
- Bayesian view
- The puzzle
- A frequentist explanation (bias, variance)
- Experiments & beyond

P

F

N

Network





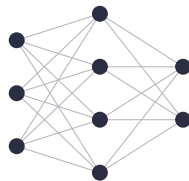
P

F

N

Fitted

Network





# P

Prior

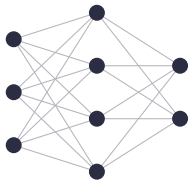
$\Pi$

# F

Fitted

# N

Network

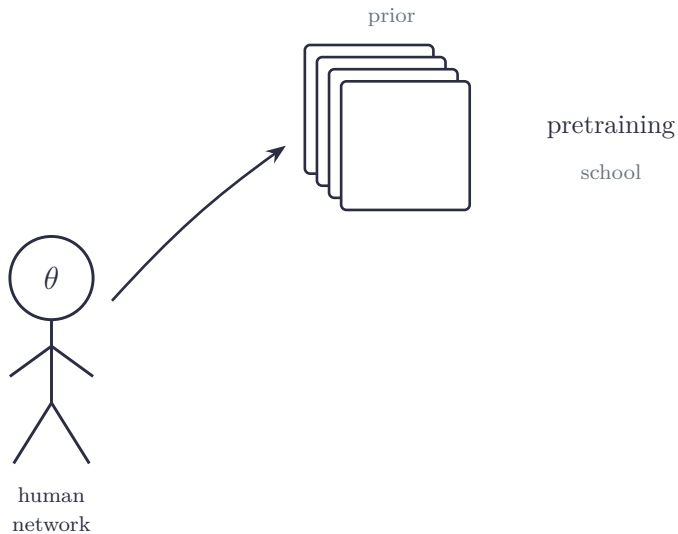


# An analogy

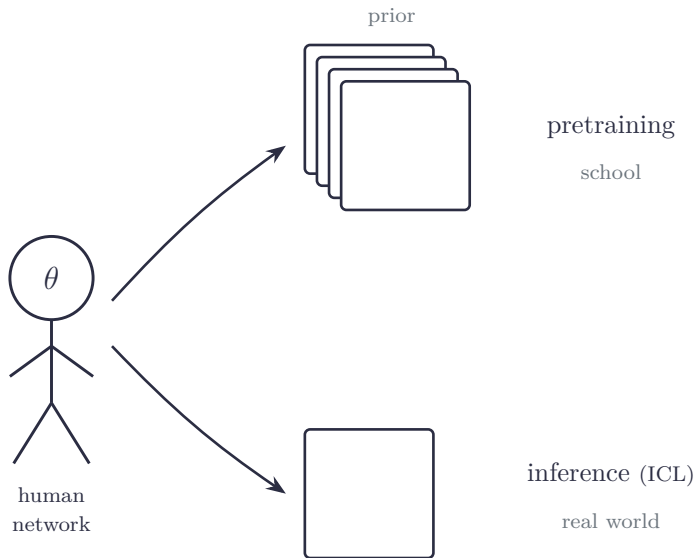


human  
network

# An analogy



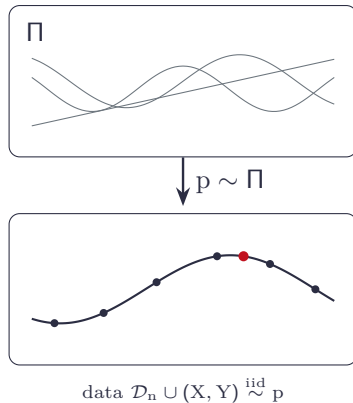
# An analogy



# The statistical model

Data is generated in two steps:

1. draw a model  $p \sim \Pi$
2. draw data  $\mathcal{D}_n = (Y_i, X_i)_{i=1}^n, (Y, X) \stackrel{\text{iid}}{\sim} p$



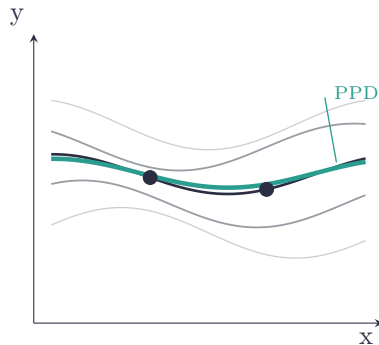
# Posterior Predictive Distribution

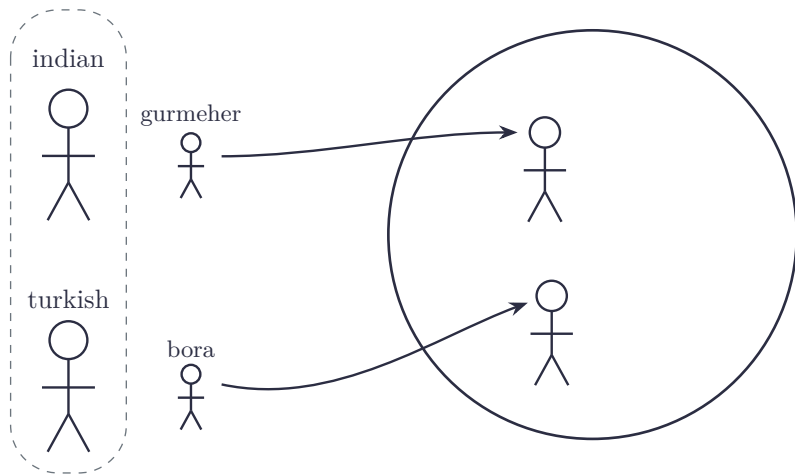
PPD averages every model,

$$\pi(y \mid x, \mathcal{D}_n) = \int p(y \mid x) d\Pi(p \mid \mathcal{D}_n)$$

weighted by its posterior

$\Pi(p \mid \mathcal{D}_n)$ : how plausible model  $p$  is, given  $\mathcal{D}_n$



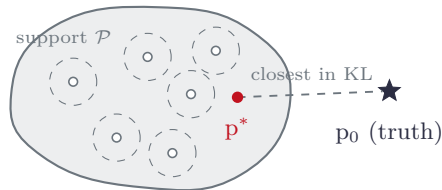


## Theorem 3.1

Under (A1)–(A2), as  $n \rightarrow \infty$ ,

$$\pi(y \mid x, \mathcal{D}_n) \longrightarrow p^*(y \mid x) \quad \text{almost surely,}$$

with  $p^*$  the KL-optimal model in  $\mathcal{P}$ .

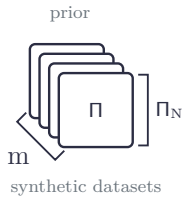


(A1) unique  $p^* \in \mathcal{P}$  minimizing  $\text{KL}(p^* \parallel p_0) < \infty$   
a reachable best target

(A2)  $\mathcal{P}$  covered by small balls of summable prior mass  
prior covers its support well

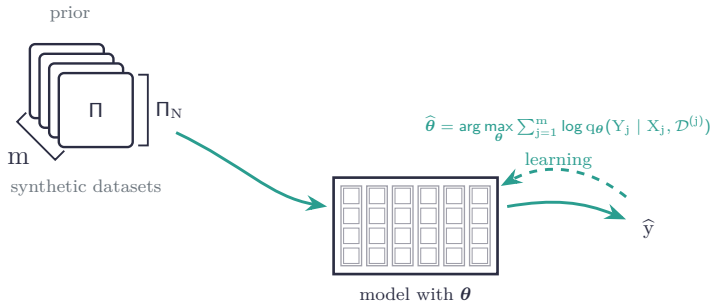
**takeaway:** more data  $\rightarrow$  best model the prior allows (even if  $p_0 \notin \mathcal{P}$ )

# PFNs pretraining & inference

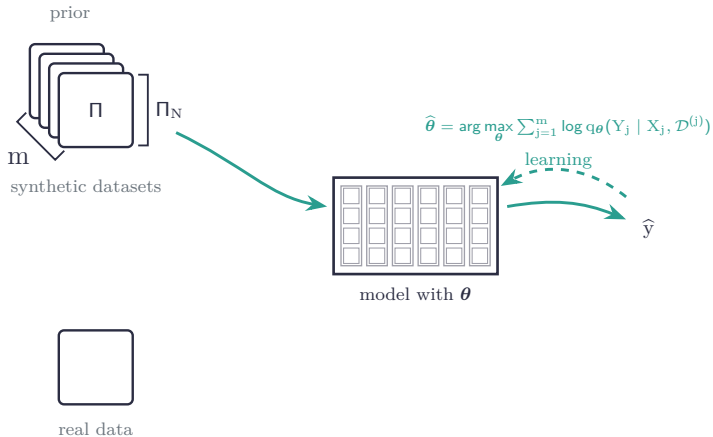


model with  $\theta$

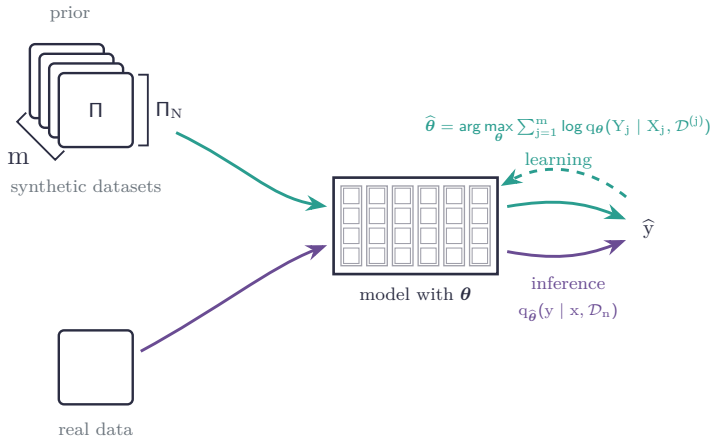
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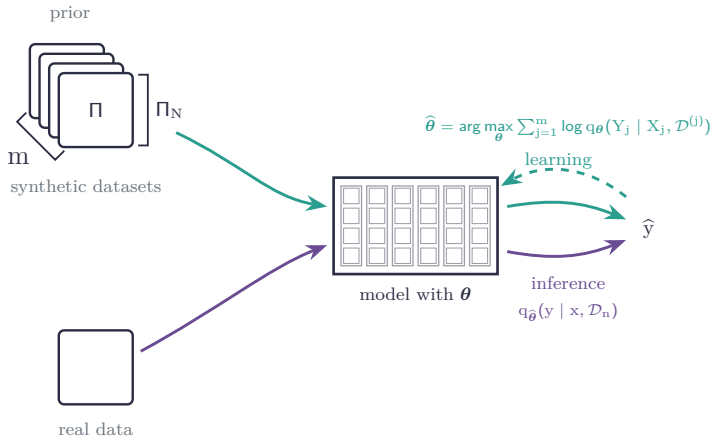
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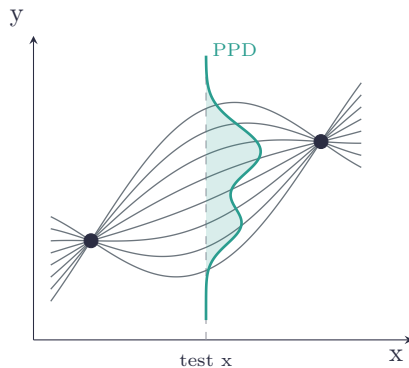


$N \sim \Pi_N$  regularizes PPD complexity (TabPFN:  $N \leq 1023$ )

$\hat{\theta} \rightarrow \theta^*$  as  $m \rightarrow \infty$  ( $\sqrt{m}$  rate)  $m$  [App. B]

# PFNs approximate PPD

- Why should you care about the PPD?
  - sum over all possible models



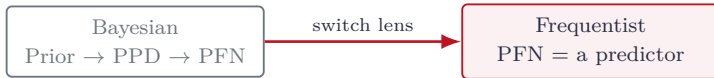
why does a PFN pre-trained on  $\leq 1000$  samples  
improve when fed larger datasets at inference??

- no reason it behaves like a PPD for unseen sizes
- Theorem 3.1 no longer applies to a frozen  $q_{\hat{\theta}}$  at large  $n$

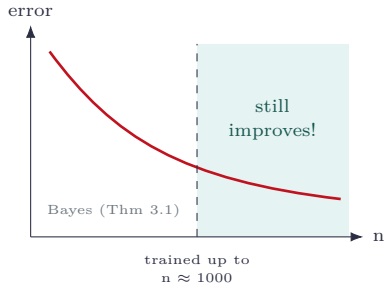
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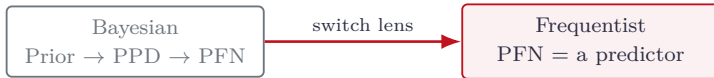


## The puzzle

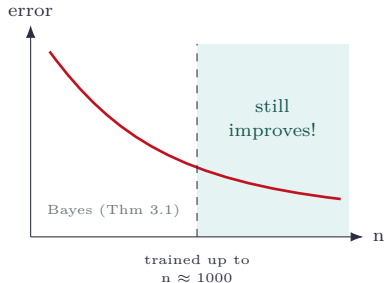


Bayes is silent past 1000.

# Why a frequentist view of PFNs?

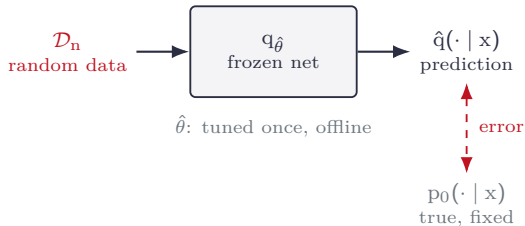


The puzzle



Bayes is silent past 1000.

The reframe: a PFN is just a predictor

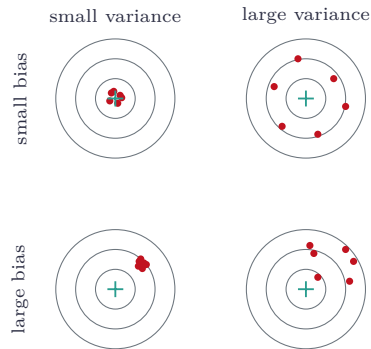


$$\text{error} = \text{bias} + \text{variance}$$

wrong on average    wobble from random  $\mathcal{D}_n$

# Bias and variance

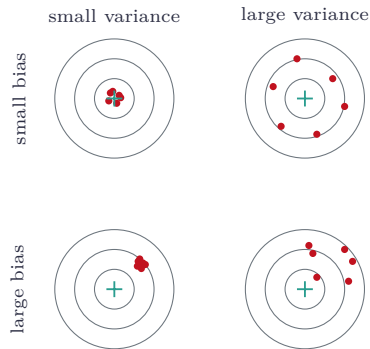
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Variance = how **scattered** the darts are.



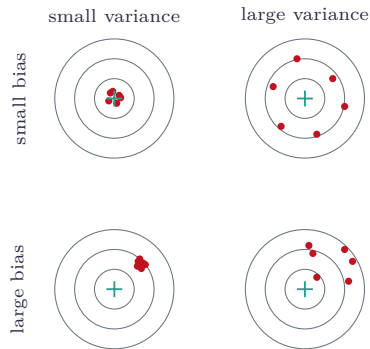
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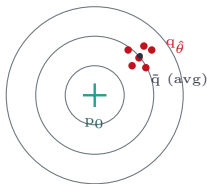
Bias = how far the darts' **centre** is from the bullseye.

**Want:** small bias and small variance (top-left).



# Bias and variance: the decomposition

$$\begin{aligned} & \underbrace{q_{\hat{\theta}}(y \mid x, \mathcal{D}_n) - p_0(y \mid x)}_{\text{error}} \\ &= \underbrace{q_{\hat{\theta}}(y \mid x, \mathcal{D}_n) - \mathbb{E}_{\mathcal{D}_n \sim p_0^n}[q_{\hat{\theta}}(y \mid x, \mathcal{D}_n)]}_{\text{variance}} \\ & \quad + \underbrace{\mathbb{E}_{\mathcal{D}_n \sim p_0^n}[q_{\hat{\theta}}(y \mid x, \mathcal{D}_n)] - p_0(y \mid x)}_{\text{bias}} \\ & \quad \bar{q} = \mathbb{E}_{\mathcal{D}_n}[q_{\hat{\theta}}] \text{ — the average prediction.} \end{aligned}$$

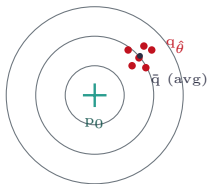


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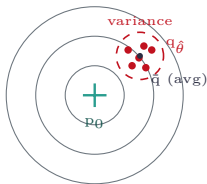
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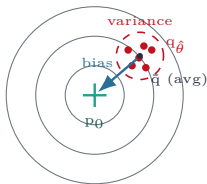
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**bias:** average  $\bar{q}$  vs the truth  $p_0$ .

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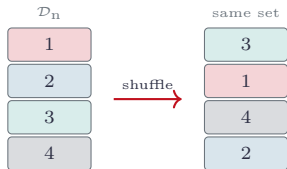
- Symmetry
- Diminishing sensitivity

Bias the offset

- Locality

A predictor is **symmetric** if shuffling the training rows never changes its answer.

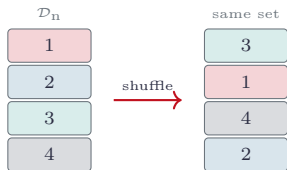
(rows are iid  $\Rightarrow$  their order is meaningless.)



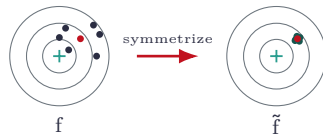
same rows, new order  $\Rightarrow$  a **symmetric**  $f$  gives the same answer.

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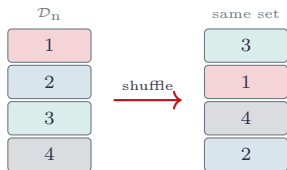
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$f$ : any predictor;  $\tilde{f}$ :  $f$  averaged over all orders.  
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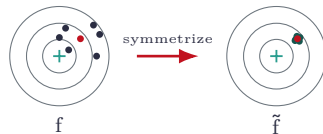
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$$\mathbb{E}[\tilde{f}] = \mathbb{E}[f]$$

same bias



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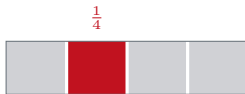
$$\text{Var}[\tilde{f}] \leq \text{Var}[f]$$

lower variance

# Diminishing sensitivity

The prediction is a **weighted average of the rows**; attention gives each row weight  $\approx \frac{1}{n}$  (the weights sum to 1).

few rows ( $n = 4$ )



each row =  $\frac{1}{4}$  (high)  $\Rightarrow$  swap one  $\Rightarrow$  **big** swing.

many rows (large  $n$ )

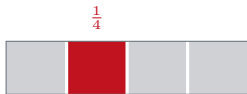


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More rows  $\Rightarrow$  each row matters **less**  $\Rightarrow$  the prediction barely changes between datasets  
 $\Rightarrow$  it is **stable**  $\Rightarrow$  **variance shrinks** (and  $\rightarrow 0$ ).

## Diminishing sensitivity — the formula

Swap one row  $\mathcal{D}_n \rightarrow \mathcal{D}'_n$ . How far can the answer move? The formula **caps** it:

$$\underbrace{\left| q_{\hat{\theta}}(\mathcal{D}_n) - q_{\hat{\theta}}(\mathcal{D}'_n) \right|}_{\text{how much the answer moved}} \leq \underbrace{L n^{-\alpha}}_{\text{a small, shrinking ceiling}}$$

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Changing a single row moves the answer by at most  $L n^{-\alpha}$  — more rows ( $n$  bigger)  $\Rightarrow$  **smaller ceiling**.

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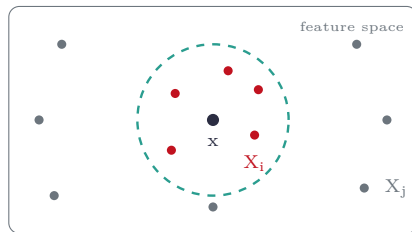
| n      | ceiling $\approx 1/n$ | swap one row moves it    |
|--------|-----------------------|--------------------------|
| 4      | 0.25                  | a <b>lot</b>             |
| 100    | 0.01                  | a little                 |
| 1000   | 0.001                 | barely                   |
| 10,000 | 0.0001                | $\approx$ <b>nothing</b> |

$\Rightarrow$  on big data one row barely matters  $\Rightarrow$  stable  $\Rightarrow$  **variance  $\rightarrow 0$** .

## Locality: only nearby points should count

Variance handled — now the bias. The tool: **locality**.

To predict at the query  $x$ , which training points should matter?



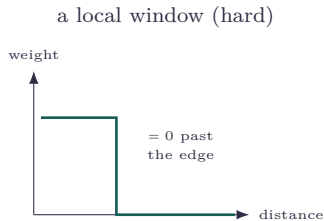
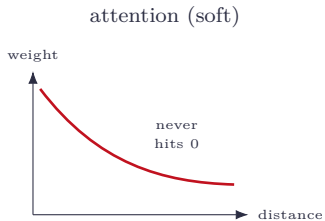
•  $x$  (the query)    • sample  $X_i$  near  $x$     • sample  $X_j$  far from  $x$  (also a sample)

Inside the ring (**near**  $x$ , red) drives the answer; the far (grey) points are ignored.

[Thm 5.4] an unbiased predictor must be **local**: flip the far-away labels  $\rightarrow$  the prediction doesn't change.

# Transformers aren't local

Attention is a **soft** weighting: it favours samples similar to  $x$ , but never drops the far ones to zero.

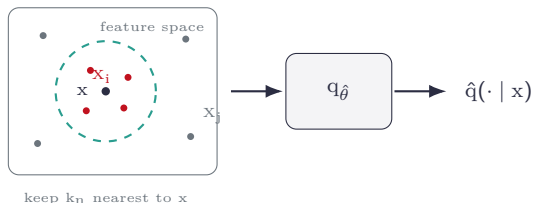


Far points **always leak in**  $\Rightarrow$  you can't flip them freely  $\Rightarrow$  **not local**.

$\Rightarrow$  the bias does not vanish — TabPFN's bias flattens out around  $n \approx 1000$  [Fig. 1].

# Localized PFNs

A simple add-on to any frozen PFN — localize at prediction time.



•  $x$  (query)    • near samples  $X_i$  (kept)    • far samples  $X_j$  (dropped)

Why it works: near  $x$  the truth is  $\approx$  flat  $\Rightarrow$  easy to fit  $\Rightarrow$  bias drops. (cost: fewer points  $\Rightarrow$  slightly more variance.)

Does it actually break the bias floor? We'll put it to the test at the end [Fig. 1].

Two more concrete predictors, each with a different bias/variance behaviour:

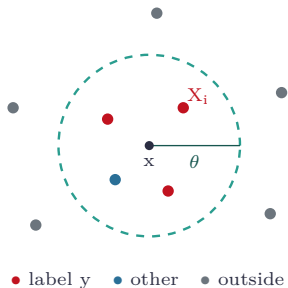
1. Window smoother
2. Classification trees

(the Localized PFN above is the third.)

# Window smoother

$$q_{\theta}(y \mid \mathbf{x}, \mathcal{D}_n) = \frac{\sum_i 1(Y_i = y) 1(\|\mathbf{X}_i - \mathbf{x}\| < \theta)}{\sum_i 1(\|\mathbf{X}_i - \mathbf{x}\| < \theta)}$$

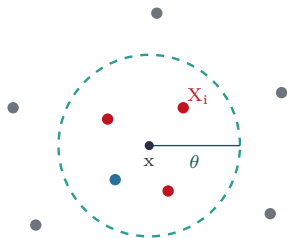
dots = training points  $\mathbf{X}_i$  (with label  $Y_i$ );  $\mathbf{x}$  = the query.



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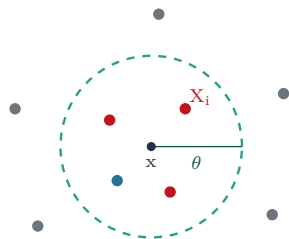
$\|\mathbf{X}_i - \mathbf{x}\| < \theta$ :  $\mathbf{X}_i$  is inside the window.



• label y   • other   • outside

# Window smoother

$$q_{\theta}(y \mid x, \mathcal{D}_n) = \frac{\sum_i \mathbf{1}(Y_i = y) \mathbf{1}(\|X_i - x\| < \theta)}{\sum_i \mathbf{1}(\|X_i - x\| < \theta)}$$



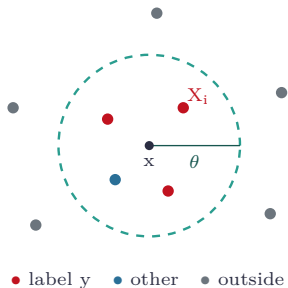
● label y   ● other   ● outside

dots = training points  $X_i$  (with label  $Y_i$ );  $x$  = the query.

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$\mathbf{1}(Y_i = y)$ : its label is  $y$  (red).

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$$q_{\theta} = \frac{\#\{\text{inside \& } y\}}{\#\{\text{inside}\}} = \frac{3}{4}.$$

## Window smoother — shrink the window, kill the bias

Now let the radius shrink with  $n$ : replace  $\theta$  by  $a_n \theta$ .

$$q_\theta(y \mid x, \mathcal{D}_n) = \frac{\sum_i 1(Y_i = y) 1(\|X_i - x\| < a_n \theta)}{\sum_i 1(\|X_i - x\| < a_n \theta)}.$$

## Window smoother — shrink the window, kill the bias

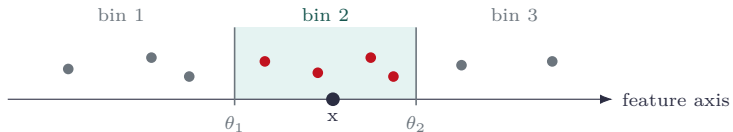
Now let the radius shrink with  $n$ : replace  $\theta$  by  $a_n \theta$ .

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Optimal schedule  $a_n = n^{-1/(4+d)}$  ( $d = \text{\#features}$ ): bias  $\rightarrow 0$  as data grows.

# Classification trees

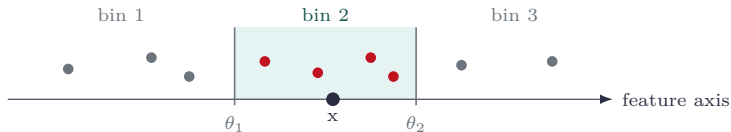
A tree cuts feature space into bins; to predict at  $x$ , report the label-mix of  $x$ 's bin.



$x$  lands in bin 2  $\Rightarrow$  use only the points there.

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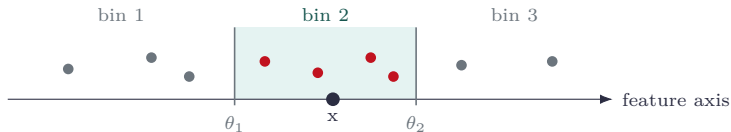
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which bin holds  $x$

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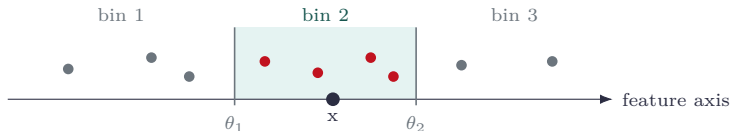
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which bin holds  $x$    the points in it   ... **with label  $y$**   $\Rightarrow$  the bin's label fraction

## Classification trees: the bias is stuck

$$\text{bias} = \underbrace{\Pr_0(Y = y \mid X \in \text{bin})}_{\text{truth averaged over the bin}} - \underbrace{p_0(y \mid x)}_{\text{truth at } x}$$

Bins are **fixed**  $\Rightarrow$  each bin blurs the same patch  $\Rightarrow$  **bias can't shrink** (more data = tighter, not truer).

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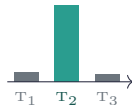
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**Fix:** average **many trees** with different splits:  $q = \sum_k w_k q_k$ .

many different trees



weight  $w_k$  = how well tree  $k$  fits  $\mathcal{D}_n$



predicts the real labels better  $\Rightarrow$  bigger weight

$n$  grows  $\Rightarrow$  best-fitting tree wins  $\Rightarrow$  bias drops.

## Experiment [Fig. 1]

The truth:  $p_0(1 | \mathbf{x}) = \frac{1}{2}(1 + \sin(\mathbf{1}^\top \mathbf{x}))$   $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_5)$ .



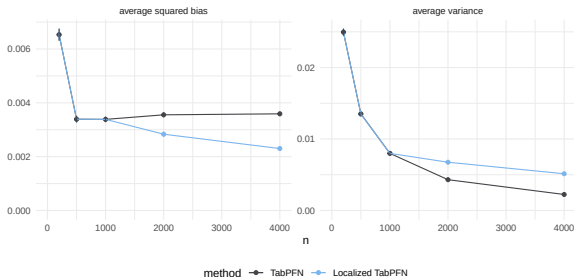
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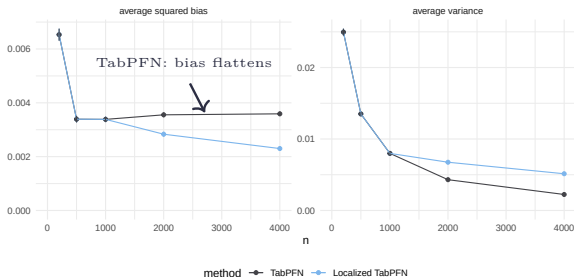


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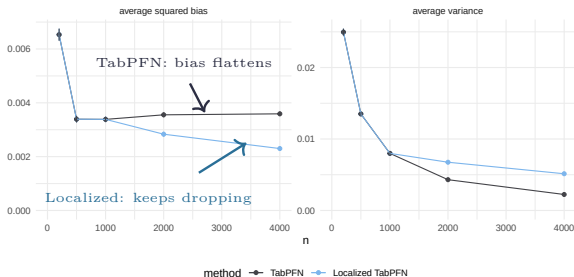


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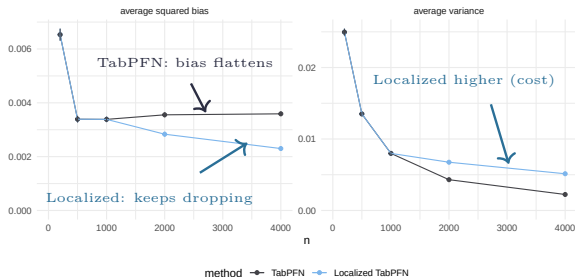


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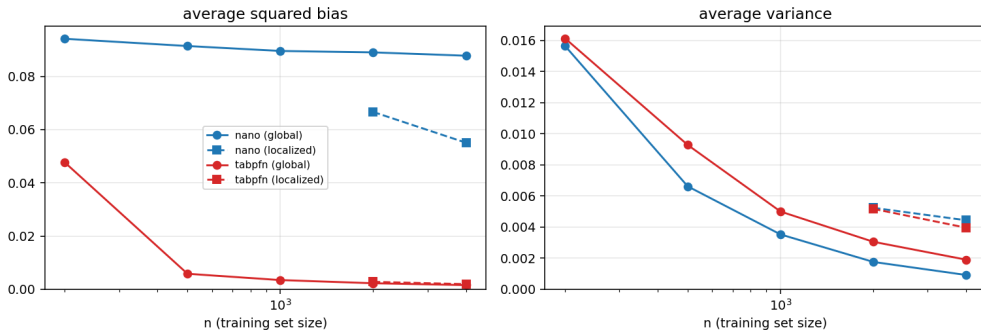
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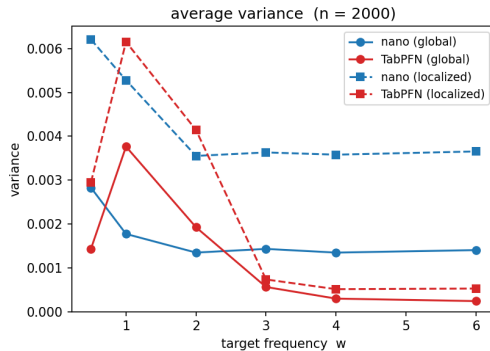
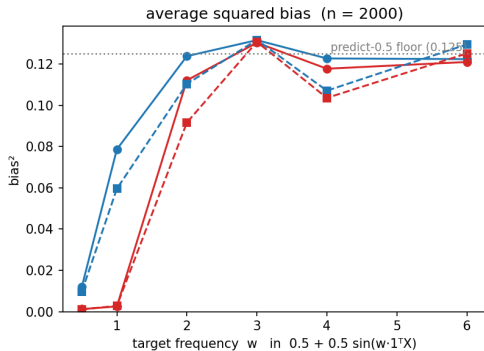
⇒ localizing breaks the bias floor, at a small variance cost.

# We redid the experiment

Nagler (2023) Fig. 1 reproduction — bias/variance vs  $n$



Frequency sweep: bias saturates as the target outpaces what the transformer can resolve



- PFNs are motivated by Bayesian ideas
- for  $n > 1000$ , a purely frequentist approach explains their behavior
- as  $n$  grows:
  - variance  $\rightarrow 0$  (transformer)
  - bias plateaus
- localize  $\Rightarrow$  bias decreases, at a small variance cost

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still so much more  
to explore!

Thank you!

Questions?

Generative model. 1. draw  $p \sim \Pi$ ; 2. draw  $\mathcal{D}_n = (Y_i, X_i)_{i=1}^n$  and  $(Y, X) \stackrel{\text{iid}}{\sim} p$ .

PPD (definition):

$$\pi(y \mid x, \mathcal{D}_n) = \Pr(Y = y \mid X = x, \mathcal{D}_n).$$

If the prior factorizes into independent  $p(y \mid x)$  and  $p(x)$  — the posterior mean:

$$\pi(y \mid x, \mathcal{D}_n) = \int p(y \mid x) d\Pi(p \mid \mathcal{D}_n).$$

A non-factorizing prior would instead give  $\pi(y \mid x, \mathcal{D}_n) = \int p(y \mid x) d\Pi(p \mid x, \mathcal{D}_n)$ .

Over all conditional densities  $q \in \mathcal{Q} = \{q : \sum_{y \in \mathcal{Y}} q(y \mid \cdot, \cdot) = 1\}$ :

$$\pi = \arg \max_{q \in \mathcal{Q}} \mathbb{E}_{\Pi}[\log q(Y \mid X, \mathcal{D}_n)].$$

Proof:  $\text{KL}(q \parallel \pi) \geq 0 \Rightarrow \mathbb{E}_{\Pi}[\log \pi] \geq \mathbb{E}_{\Pi}[\log q]$ .

Parametric KL-optimal target (averaged over sizes  $N \sim \Pi_N$ ) and its Monte-Carlo estimate:

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\Pi_N} \mathbb{E}_{\Pi}[\log q_{\theta}(Y \mid X, \mathcal{D}_N)],$$

$$\hat{\theta} = \arg \max_{\theta} \sum_{j=1}^m \log q_{\theta}(Y_j \mid X_j, \mathcal{D}^{(j)}).$$

Support  $\mathcal{P} = \{p : \Pi(p) > 0\}$ . Under (A1)–(A2), as  $n \rightarrow \infty$ ,

$$\pi(y \mid x, \mathcal{D}_n) \rightarrow p^*(y \mid x) \quad \text{a.s.}, \quad p^* = \arg \min_{p \in \mathcal{P}} \text{KL}(p^* \parallel p_0).$$

(A1) unique  $p^* \in \mathcal{P}$  with  $\text{KL}(p^* \parallel p_0) < \infty$ .

(A2) for every  $\alpha \in (0, \frac{1}{2})$  there are  $B_1, \dots, B_{J(\alpha)}$  with

$$\mathcal{P} \subseteq \bigcup_j B_j, \quad \sup_{p, p' \in B_j} H(p, p') \leq 4(\alpha^2/2)^{1/\alpha}, \quad \sum_j \Pi(B_j)^\alpha < \infty.$$

Key step (de Blasi–Walker '13):  $\Pi_n\{p : H(p, p^*) > \epsilon\} \rightarrow 0$ , with  $\Pi_n(A) = \int_A d\Pi(p \mid \mathcal{D}_n)$ .

Size-specific optimum  $\boldsymbol{\theta}_n^* = \arg \max_{\boldsymbol{\theta}} \mathbb{E}_{\Pi}[\log q_{\boldsymbol{\theta}}(Y \mid X, \mathcal{D}_n)]$ ; overall  $\widehat{\boldsymbol{\theta}} = \boldsymbol{\theta}^* + O_p(m^{-1/2})$ .

Estimator.  $\widehat{\boldsymbol{\theta}} \rightarrow \boldsymbol{\theta}^*$  in probability, and

$$\sqrt{m}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) \rightarrow \mathcal{N}(0, \boldsymbol{\Sigma}_{\boldsymbol{\theta}^*}), \quad \boldsymbol{\Sigma}_{\boldsymbol{\theta}^*} = \mathbf{I}_{\boldsymbol{\theta}^*}^{-1} \mathbf{V}_{\boldsymbol{\theta}^*} \mathbf{I}_{\boldsymbol{\theta}^*}^{-1},$$

$$\mathbf{I}_{\boldsymbol{\theta}^*} = \mathbb{E}_{\Pi_N} \mathbb{E}_{\Pi}[\nabla_{\boldsymbol{\theta}}^2 \log q_{\boldsymbol{\theta}^*}], \quad \mathbf{V}_{\boldsymbol{\theta}^*} = \mathbb{E}_{\Pi_N} \mathbb{E}_{\Pi}[\nabla_{\boldsymbol{\theta}} \log q_{\boldsymbol{\theta}^*} \nabla_{\boldsymbol{\theta}}^{\top} \log q_{\boldsymbol{\theta}^*}].$$

Predictor.  $\sqrt{m}(q_{\widehat{\boldsymbol{\theta}}} - q_{\boldsymbol{\theta}^*}) \Rightarrow \mathcal{G}$  (mean-zero Gaussian process),

$$\text{Var}(q_{\widehat{\boldsymbol{\theta}}}(y \mid x, \mathcal{D}_n)) \approx \frac{1}{m} \nabla_{\boldsymbol{\theta}} q_{\boldsymbol{\theta}^*}^{\top} \boldsymbol{\Sigma}_{\boldsymbol{\theta}^*} \nabla_{\boldsymbol{\theta}} q_{\boldsymbol{\theta}^*}.$$